Adrien Trahan

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Research Interests

Reinforcement learning, representation learning, and learning-guided search. I am particularly interested in predictive and self-supervised representations (JEPA-style architectures) and in how learned representations and value functions can guide planning and reasoning. I am also open to other directions in machine learning and mathematics.

Education

Polytechnique Montréal — **August 2023 to April 2027**
Bachelor's Degree in Software Engineering — Director General's Excellence Distinction – GPA: 3.84 / 4.0
Concentration in Artificial Intelligence and Data Science
Relevant courses: Machine Learning, Reinforcement Learning, Optimization Methods and Algorithms, Probability and Statistics Methods for AI, Applied Linear Algebra

Politecnico di Torino (Turin, Italy) — **September 2025 to June 2026**
2-semester exchange, registered at the Master's level in Computer Engineering — 29.08 / 30

Research Experience

Task Arithmetic and Federated Learning – Research Project — **2026**
Politecnico di Torino — github.com/AdrienTrahan/FL-task-arithmetic

- Studied task arithmetic in a federated learning setting and wrote up the findings as a research paper
- Ran large-scale experiments with pretrained vision models such as DINO, totalling 60+ GPU hours

Independent Research — **present**
Self-directed

- Small-scale experiments with JEPA-style architectures and regularization approaches (sub-JEPA, SIGReg, VISReg) for self-supervised representation learning in MiniGrid environments
- Exploring whether DeepCubeA-style learned heuristic search can be applied to automated theorem proving, framing proof construction as pathfinding in a graph of proof states

Professional Experience

Ivado Labs — **May 2025 to August 2025**
Software Engineering Intern (machine learning / data)

- Contributed to the outperformance of a trading model on a $50M+ portfolio on the electricity market by identifying and integrating key new data sources
- Improved the reliability of the model's predictions by uncovering flaws in their distribution and building a data (ETL) pipeline from scratch in Python

Thirdbridge Inc. — **May 2024 to August 2024**
Software Development Intern

- Delivered the redesign of a $1M+ contracting company's website, still in production over 2 years later (React, GatsbyJS, Payload CMS)

L3Harris Technologies, Inc. — **May 2022 to August 2022, May 2023 to August 2023**
Business Intelligence Intern (returning intern)

- Improved work-progress tracking for 10 teams via Power BI dashboards, coordinating the SQL integration of hundreds of tables from 10+ databases

Awards and Distinctions

Computer Science Games (CS Games) — **April 2024**
First place in an inter-university competition — 14 universities, nearly 300 students

Hackatown, Montréal — **January 2020**
Judges' Favorite Award among 50 university teams — $800 in prizes

Polytechnique Hackathon — **April 2017 and April 2019**
First place in programming contests — $1,000 in scholarships

Technical Skills

Proficient: PyTorch, Python, Typescript, React, SvelteKit, Swift, MongoDB
Experienced: C++, Java, PostgreSQL, Microsoft SQL Server, Git